LEVEL UP ACADEMY

Class 10 | CBSE | Science

Pre-Board Mock 3 (CBSE pattern)

Chapter / Topic: Full syllabus (Chapters 1–13)

Name: _____ Roll No.: _____ Date: _____

Time: 3 hours **Max. Marks: 80**

General instructions:

1. This paper has 39 questions in 5 sections (A–E). All questions are compulsory. Internal choice is given in some questions.
2. Section A: Q1–Q20, 1 mark each (MCQs and Assertion–Reason). Section B: Q21–Q26, 2 marks each. Section C: Q27–Q33, 3 marks each.
3. Section D: Q34–Q36, 5 marks each. Section E: Q37–Q39, case-based, 4 marks each.
4. Write balanced equations, show formulae and units in numericals, and draw neat labelled diagrams wherever required.
5. This mock follows the reported 2026-27 pattern. Check the official CBSE sample paper for any changes.

SECTION A

20 marks

Q1–Q16: choose the correct option. Q17–Q20: Assertion–Reason. For these, choose (a) both A and R true and R explains A; (b) both true but R does not explain A; (c) A true, R false; (d) A false, R true.

Q1. Which of the following is NOT a redox reaction? **[1]**

(a) $CuO + H_2 \rightarrow Cu + H_2O$ (b) $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$
(c) $2K + Cl_2 \rightarrow 2KCl$ (d) $BaCl_2 + H_2SO_4 \rightarrow BaSO_4 + 2HCl$

Q2. Which of these salts does NOT contain water of crystallisation? **[1]**

(a) Blue vitriol (b) Baking soda
(c) Washing soda (d) Gypsum

Q3. An aqueous solution has a pH of 0. It is: **[1]**

(a) strongly acidic (b) strongly alkaline
(c) neutral (d) amphoteric

Q4. Which property of metals makes them suitable for making bells? **[1]**

(a) Ductility (b) Malleability
(c) Lustre (d) Sonority

Q5. How many covalent bonds are there in a molecule of methane? **[1]**

(a) 1 (b) 2
(c) 4 (d) 5

Q6. The two carbon atoms in ethyne are joined by: **[1]**

(a) a triple bond (b) a single bond
(c) a double bond (d) an ionic bond

Q7. In a plant cell, photosynthesis takes place in the: **[1]**

(a) mitochondria (b) chloroplasts
(c) nucleus (d) ribosomes

Q8. Lymph is: **[1]**

(a) a colourless fluid that also carries digested and absorbed fat
(b) a red fluid that carries oxygen
(c) a fluid made in the kidney
(d) a fluid found only in plants

Q9. Iodine is needed by the body to make: **[1]**

(a) insulin (b) adrenaline
(c) testosterone (d) thyroxin

Q10. Which of the following involves sexual reproduction? **[1]**

(a) Budding in yeast
(b) Pollination followed by fertilisation in mustard
(c) Fragmentation in Spirogyra
(d) Regeneration in Planaria

Q11. Two heterozygous tall pea plants (Tt × Tt) are crossed. What fraction of the offspring will be tall? **[1]**

(a) 1/4 (b) 1/2
(c) 3/4 (d) 1

Q12. Which of these is an artificial ecosystem? **[1]**

(a) Lake (b) Aquarium
(c) Forest (d) Grassland

Q13. Which mirror do dentists use to see enlarged images of teeth? **[1]**

(a) Plane mirror (b) Convex mirror
(c) Any mirror (d) Concave mirror

Q14. 1 dioptre is the power of a lens whose focal length is: **[1]**

(a) 1 cm (b) 10 cm
(c) 1 m (d) 10 m

Q15. The potential difference across a resistor is measured with a: **[1]**

(a) ammeter connected in series
(b) voltmeter connected in series
(c) voltmeter connected in parallel
(d) ammeter connected in parallel

Q16. The direction of the magnetic field around a straight current-carrying conductor is given by: **[1]**

(a) the right-hand thumb rule (b) Fleming's left-hand rule
(c) Ohm's law (d) Snell's law

Q17. **Assertion (A):** Distilled water does not conduct electricity well, but tap water does.
Reason (R): Tap water contains dissolved salts, which provide ions. **[1]**

Q18. **Assertion (A):** Human beings have a four-chambered heart.
Reason (R): Fish have a two-chambered heart. **[1]**

Q19. **Assertion (A):** Concave mirrors are used in solar furnaces.
Reason (R): A concave mirror concentrates sunlight at its focus. **[1]**

Q20. **Assertion (A):** Diamond is a good conductor of electricity.
Reason (R): Graphite is a good conductor of electricity. **[1]**

SECTION B

12 marks

Q21. Hydrogen sulphide gas is passed through copper sulphate solution and a black precipitate forms. Write the balanced equation and name the type of reaction. **[2]**

Q22. Give two differences between the chemical properties of metals and non-metals. **[2]**

OR

Write the chemical formulae of (a) baking soda, (b) washing soda, (c) bleaching powder, (d) Plaster of Paris.

Q23. Name the end products of the digestion of (a) carbohydrates, (b) proteins, (c) fats. **[2]**

Q24. Why are offspring produced by asexual reproduction almost identical to their parent? **[2]**

OR

Why is variation important for the survival of a species?

Q25. A convex lens of focal length 50 cm is placed in contact with a concave lens of focal length 20 cm. Find the power of the combination. Is it converging or diverging? **[2]**

Q26. Bulbs rated 60 W and 100 W (both for 220 V) are connected in series across 220 V. Which one glows brighter? Explain. **[2]**

OR

Give two differences between series and parallel combinations of resistors.

SECTION C

21 marks

Q27. A compound X reacts with sodium to give hydrogen gas. When X is heated with excess concentrated sulphuric acid at 443 K, it gives an unsaturated compound Y. Identify X and Y, and write the balanced equations for both reactions. **[3]**

Q28. Write balanced equations for the reactions of: **[3]**

(i) zinc with sodium hydroxide solution **[1]**

(ii) aluminium oxide with sodium hydroxide **[1]**

(iii) copper with oxygen on heating **[1]**

Q29. How do plants respond to stimuli without a nervous system? Explain with the examples of the touch-me-not plant and a climbing tendril. **[3]**

Q30. Give the function of each gland: (i) pituitary, (ii) thyroid, (iii) adrenal. **[3]**

OR

Why is the use of iodised salt advised? Name the gland involved and the disease caused by a deficiency.

Q31. State the 10 per cent law. If producers in a food chain trap 1,00,000 J of energy, how much energy reaches the fourth trophic level? Why are top carnivores few in number? **[3]**

Q32. A 10 cm tall object is placed 20 cm from a convex lens of focal length 30 cm. Find the position, size and nature of the image. **[3]**

OR

A concave mirror of focal length 20 cm forms a real image 60 cm from the mirror. Find the object distance and the magnification.

Q33. An electric iron uses 840 W at its highest setting and 360 W at its lowest, on a 220 V supply. Find the current and the resistance in each case. **[3]**

SECTION D

15 marks

Q34. Answer the following: **[5]**

(i) How do acids react with metal carbonates and metal hydrogencarbonates? Give one balanced equation for each, and describe the test for the gas formed. **[3]**

(ii) Why should curd not be stored in copper or brass vessels? **[1]**

(iii) What is a universal indicator? **[1]**

OR

(a) Explain, with one balanced equation each, the combustion, oxidation, addition and substitution reactions of carbon compounds. [4]
(b) Why do unsaturated hydrocarbons burn with a sooty flame? [1]

Q35. Answer the following: **[5]**

(i) Describe sexual reproduction in flowering plants, from pollination to the formation of seed and fruit. **[3]**

(ii) Give two differences between self-pollination and cross-pollination. **[2]**

OR

(a) Describe the role of the liver and the pancreas in digestion. [3]
(b) How are villi suited to absorbing food? [2]

Q36. Answer the following: **[5]**

(i) Explain the refraction of light through a rectangular glass slab. What is lateral displacement? **[3]**

(ii) The refractive index of water is 1.33. Find the speed of light in water. ($c = 3 \times 10^8$ m/s) **[2]**

OR

(a) What is dispersion of light? Explain how a rainbow is formed. [3]
(b) Why are danger signals red? [2]

SECTION E

12 marks

Read each passage and answer the questions that follow.

Q37. Doctors use Plaster of Paris to support fractured bones in the right position. It is a white powder made by heating gypsum carefully at 373 K. When mixed with water, it changes back into gypsum and sets into a hard, solid mass. **[4]**

(i) Give the chemical name and formula of Plaster of Paris. **[1]**

(ii) How is it prepared? **[1]**

(iii) Why must Plaster of Paris be stored in a moisture-proof container? Give one other use. **[2]**

Q38. Mendel crossed pea plants that produced yellow seeds (YY) with plants that produced green seeds (yy). All the F_1 seeds were yellow. When the F_1 plants were self-pollinated, he obtained 6022 yellow seeds and 2001 green seeds in the F_2 generation. **[4]**

(i) Which seed colour is dominant? **[1]**

(ii) What is the approximate ratio of yellow to green seeds in F_2? **[1]**

(iii) What are the possible genotypes of the yellow F_2 seeds, and what fraction of the yellow seeds are pure-breeding? **[2]**

Q39. In scrapyards, large electromagnets hanging from cranes lift heavy pieces of scrap iron. An electromagnet is a solenoid with a soft iron core. When the current is switched on, the iron core becomes a strong magnet. When the current is switched off, it loses its magnetism. **[4]**

(i) What is an electromagnet? **[1]**

(ii) Why is soft iron used for the core? **[1]**

(iii) Give two ways to make the electromagnet lift heavier loads. What happens to the scrap when the current is switched off? **[2]**

~ All the best for your boards! ~